

My original widget for the independent study is one that recognizes differences in the topics of documents based on the universities they were provided by. I stepped away from differentiating based on university provided by because the worksets were becoming too big for topic models to handle. Now, I'm focusing on the differences based on publication location. I'm testing this with a set of documents during a period of time during the Vietnam war (1967 – 1970).

I have cleaned the dataset of stopwords and nonwords, separated the workset's volumes based on publication location, and fed each to the LdaModel from the Gensim library, comparing the results to the HathiTrust InPhO Topic Explorer. The code currently prints out the top 5 topics of relevance to the set of documents per state, as well as the top 20 words per topic so that we can understand the distinguishing topics written about in the news at that time more.

An issue I'm running into is that certain volume IDs are returning a 404 error despite the fact that, at least on the workset builder, the volume should be pullable, so sadly I'm not able to pull as large of a dataset as I would like to test on.

Currently I'm not using a specific visualization. I am showing each location's top 5 topics, as well as the 20 words and probabilities associated with each topic. With this, I'm hoping to see the differences between topics of interest in different states during this time period of conflict.

For future work, I would like to make this more scalable so that it can be any set of volumes that is being analyzed. As per the raw jupyter notebook, it's not difficult to just change the workset id and get the results of this process for up to 1000 of the volumes in that workset, so making it more accessible and with clean UI could be a next step. I also think it would be

interesting to look at the changes of topics of interest per year or time period in a specific region, so that you can recognise the changes in political and socioeconomic climates in a region, if any.

Github repo: <https://github.com/HaasikaP/HathiTrust-TorchliteProject>